

Shock wavefronts for parabolic equations with sign-changing diffusivity

joint work with Andrea Corli (Ferrara, IT) and Luisa Malaguti (Modena & Reggio Emilia, IT)

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AIMS 2026

Nonlinear differential equations: control, delay, and boundary value problems

10 July 2026, Athens

The general problem

Consider, for $u \in [0, 1]$, the **diffusion-reaction scalar parabolic equation**

$$u_t = (D(u)u_x)_x + g(u),$$

or, in *conservative form*,

$$u_t = P(u)_{xx} + g(u) \quad \text{where} \quad P'(u) = D(u). \quad (1)$$

where D stands for the **diffusivity**, g for the **reaction**, and P for the **potential**.

General framework:

($\mathcal{R}$) $P \in C^2([0, 1])$, $g \in C^0([0, 1])$; both P' and g have a finite number of zeros.

(Motivations: [Johnston-Baker-McElwain-Simpson, *Sci. Rep.* \(2017\)](#) on modeling co-operation, competition and crowding in biology.)

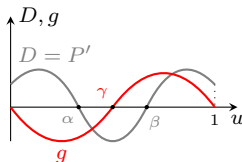
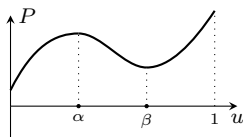
Case study assumptions ($\mathcal{S}$):

for some

$$0 < \alpha < \gamma < \beta < 1,$$

assume that

$$P' > 0 \text{ in } (0, \alpha) \cup (\beta, 1) \text{ and } P' < 0 \text{ in } (\alpha, \beta);$$
$$g(0) = g(1) = 0, \quad g < 0 \text{ in } (0, \gamma) \text{ and } g > 0 \text{ in } (\gamma, 1).$$



We look for (decreasing) wavefronts (WFs for short): $u(x, t) = \varphi(\xi)$ with $\xi = x - ct \in \mathbb{R}$ such that

$$\varphi(-\infty) = 1 \quad \text{and} \quad \varphi(\infty) = 0.$$

The equation satisfied by the profile of a (regular) WF is

$$P(\varphi)'' + c\varphi' + g(\varphi) = 0 \quad \text{in } \mathbb{R}$$

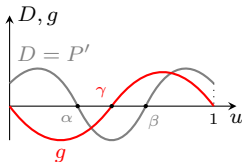
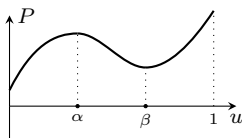
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Short recap about regular WFs

- $D > 0$ in $(0, 1)$: classical results (Fife (1979), Gilding-Kersner (2004));
- D changes sign and g monostable (Maini-Malaguti-Marcelli-Matucci (2007), B.-Corli-Malaguti (2021),...);
- D changes sign and g bistable (Maini-Malaguti-Marcelli-Matucci (2007), Kuzmin-Ruggerini (2011), B.-Corli-Malaguti (2022),...);

In particular, in Kuzmin-Ruggerini is proved that, under our notations, WFs exist only if

$$\text{either } \gamma < \alpha \quad \text{or} \quad \gamma > \beta,$$

that is if γ lies in regions where $D > 0$.

$\Rightarrow$ (regular) WFs does not exist when $\alpha < \gamma < \beta$.

About our problem:

- Numerical simulations $\Rightarrow$ shock WFs skipping the whole interval (α, β) (Johnston et al. (2017));
- Shock WFs can be selected as limits of augmented equations (Li-van Heijster-Marangell-Simpson (2020-21), Witelski (1995)) of the type

$$u_t = P(u)_{xx} + g(u) - \varepsilon^2 u_{xxxx}, \quad u_t = P(u)_{xx} + g(u) + \varepsilon^2 u_{xxt}.$$

Discontinuous solutions to flux-saturated diffusion equations:

- Entropy solutions: Bertsch-Dal Passo, Caselles et al.;
- For WFs: Calvo-Campos-Caselles-Sanchez-Soler, Caselles, also Rosenau et al, Kurganov (1997-2005), Campos-Corli-Malaguti(2021).

Our contribution

In case of (S) , in B.-Corli-Malaguti, preprint, we study and provide fine results for shock WFs, from a **distributional notion of shock traveling wave** defined directly for the original equation, without regularization.

Our approach under $(\mathcal{R})$

- Assume $(\mathcal{R})$.
- Let $I \subset \mathbb{R}$ be an interval.
 - Let $\varphi : I \rightarrow [0, 1]$ be continuous, except possibly at finitely many jump points.

A function $u(x, t) = \varphi(x - ct)$ is a *traveling-wave* solution of (1) if

$$\int_I P(\varphi) \psi'' d\xi = c \int_I \varphi \psi' d\xi - \int_I g(\varphi) \psi d\xi, \quad \psi \in C_c^\infty(I).$$

We call u :

- a **regular TW** if φ is continuous;
- a **shock TW** if φ has jump discontinuities.

Consequences of the definition

Suppose φ is shock with jump point ξ_0 , where φ has side limits

$$\varphi_\ell := \varphi(\xi_0^+) \quad \text{and} \quad \varphi_r := \varphi(\xi_0^-):$$

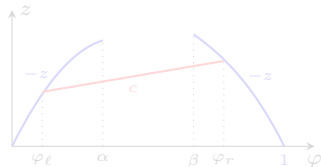
$$P(\varphi_\ell) = P(\varphi_r) \quad \text{and} \quad c = \frac{D(\varphi_\ell)\varphi'(\xi_0^+) - D(\varphi_r)\varphi'(\xi_0^-)}{\varphi_r - \varphi_\ell}$$

The former can be read as an **equal-area** condition, as

$$\int_{\varphi_\ell}^{\varphi_r} D = 0,$$

and implies that φ_ℓ and φ_r cannot belong to the same “connected component of monotonicity” of $P \Rightarrow$ there could be a number of jumps up to the changes of sign of D .

The latter is a sort of **Rankine-Hugoniot condition** in terms of $P(\varphi)'(\xi_\sigma) = z(\sigma)$.



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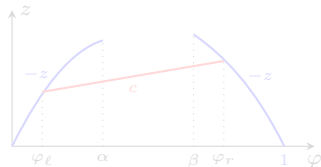
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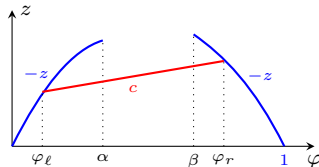
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The main Theorem under $(\mathcal{S})$

Theorem 1

Under $(\mathcal{S})$:

- (1) $P(1) > P(0) \Rightarrow$ one-parameter family of **shock WFs** with speed c and a **unique jump**, from φ_ℓ to φ_r :

$$[0, 1] \ni s \mapsto [c(s), \varphi_\ell(s), \varphi_r(s)],$$

such that $\varphi_\ell(s)$ and $\varphi_r(s)$ are strictly increasing;

- $c(s)$ is continuous and strictly decreasing^a;
- no other shock WFs.

- (2) $P(1) = P(0) \Rightarrow$ **only** the following **piecewise constant shock WFs**:

- one jump, from 1 to 0, with speed $c = 0$;
- (if also $P(\gamma) = P(0)$) two jumps, from 1 to γ and from γ to 0; the speed is again 0;

- (3) $P(1) < P(0) \Rightarrow$ no shock WFs.

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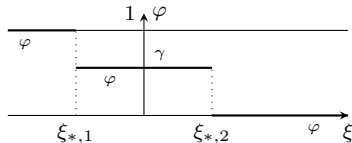
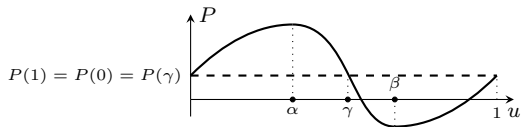
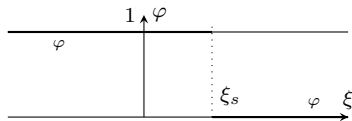
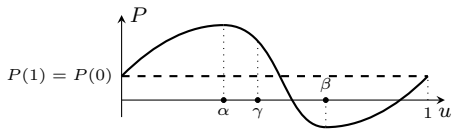
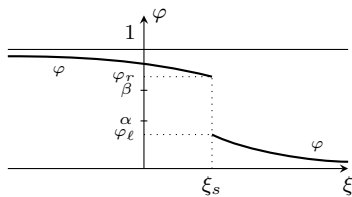
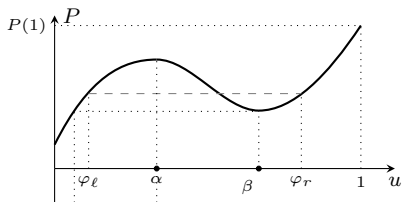
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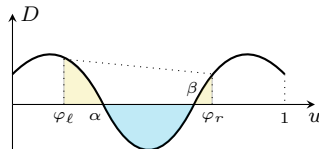
- Recall: $P(1) - P(0) = \int_0^1 D(\sigma) d\sigma$.



Sketches of the proof of Theorem 1 - Preliminary ingredients

- If $(S) \Rightarrow$ a shock WF admits a **unique jump**

$$\varphi_\ell \in [0, \alpha] \quad \text{and} \quad \varphi_r \in (\beta, 1].$$



- Hence, from monotonicity of P :

$$P(1) - P(0) \geq P(\varphi_r) - P(\varphi_\ell) = 0,$$

and

$$P(1) = P(0) \quad \Longleftrightarrow \quad \varphi_\ell = 0 \quad \text{and} \quad \varphi_r = 1.$$

- If $P(1) > P(0)$,

$$(\varphi_\ell, \varphi_r) = (\varphi_\ell, \eta(\varphi_\ell)) \quad \text{for } \varphi_\ell \in [a, b] \subset [0, \alpha]$$

and $\eta : [a, b] \rightarrow [\beta, 1]$ of class $C^0([a, b]) \cap C^1(\text{int}([a, b]))$ such that $\eta' > 0$.

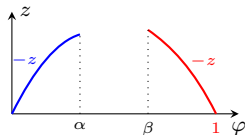
Sketches of the proof of Theorem 1 - The first-order singular reduction

- For every $c \in \mathbb{R}$ there exist **regular** profiles of **semi-WFs** connecting, resp., 1 to β and α to 0.
- In $(0, \alpha) \cup (\beta, 1)$, we define

$$z_c(\sigma) := D(\sigma)\varphi'(\varphi^{-1}(\sigma)) \neq 0, \quad \sigma \in (0, \alpha) \cup (\beta, 1),$$

which satisfies

$$\begin{cases} \dot{z}_c(\sigma) = -c - \frac{D(\sigma)g(\sigma)}{z_c(\sigma)}, & \sigma \in (0, \alpha), \\ z_c(\sigma) < 0, & \sigma \in (0, \alpha), \\ z_c(0) = 0, \end{cases} \quad \text{and} \quad \begin{cases} \dot{z}_c(\sigma) = -c - \frac{D(\sigma)g(\sigma)}{z_c(\sigma)}, & \sigma \in (\beta, 1), \\ z_c(\sigma) < 0, & \sigma \in (\beta, 1), \\ z_c(1) = 0. \end{cases}$$



Sketches of the proof - (RH) condition

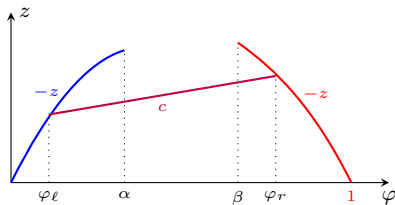
The (RH) condition becomes

$$c = \frac{z_c(\varphi_\ell) - z_c(\eta(\varphi_\ell))}{\eta(\varphi_\ell) - \varphi_\ell}$$

$\iff c \in \mathbb{R}$ and $\varphi_\ell \in [a, b]$ are such that

$$\mathcal{F}(c, \varphi_\ell) := z_c(\varphi_\ell) - z_c(\eta(\varphi_\ell)) - c(\eta(\varphi_\ell) - \varphi_\ell) = 0.$$

$$\Rightarrow c = c(\varphi_\ell).$$



- **Let $c^* \in \{c(s) : s \in [0, 1]\}$:** both cases $c^* > 0$ and $c^* < 0$ can occur.
- Thanks to fine estimates on threshold wave speeds for regular TWs, we can give estimates on the interval $\{c(s) : s \in [0, 1]\}$.
- **Uniqueness:** To single out the profile (or, equivalently, the speed c^*): choose the minimum speed c^* .

When selecting the minimal-speed wavefront, its corresponding profile remains in $[0, \alpha]$ as long as possible, and jumps otherwise.

- In B.-Corli-Malaguti, *Shock wavefronts with diffusivity changing sign once*, we propose a systematic study on the occurrence of shock WFs and their relations with the known continuous WFs, in the cases

D changes sign once.

- In B.-Corli-Trullo, we are addressing the case of shock WFs when a

convective term

is added to the model.

Thank you for your attention